



NCIE-006

Memory & Context Architecture Specification

In development — for human review — Version 1.1

Consolidates persistent Memory and Context architecture: classes, temporal integrity, provenance, authorization and Institutional Knowledge lifecycle

Model-independent: Memory survives ARGUS model/provider replacement and does not depend on any conversation-history feature

Traces to NCIE-002 v0.4 (architecture), NCIE-003 v0.3 (canonical data), NCIE-004 v1.1 (engineering) and NCIE-005 v1.2 (ARGUS behavior)

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Document Governance

Document	NCIE-006 Memory & Context Architecture Specification
Edition	In development — Version 1.1
Baseline authorized	NCIE-006 Skeletal v0.1 — approved for full expansion; 36-chapter structure locked
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.2
Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.4
Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.3 (Canonical Data Architecture Baseline)
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.1 (Approved Engineering Baseline)
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.2
Owner	National Communications Intelligence Ecosystem (NCIE) Architecture Function
Reviewer	David King Boison (PhD) — Academic Intelligence Center
Status	IN DEVELOPMENT — FOR HUMAN REVIEW until explicitly approved

0.1 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture. NCIE-006 implements and consolidates Memory/Context requirements established upstream; it does not reinterpret them, and it does not duplicate NCIE-007 (AI/Agent Orchestration), NCIE-009 (Security/IAM), NCIE-010 (Evidence/Provenance/Audit), NCIE-011 (API/Integration), NCIE-014 (Database & Storage Engineering) or NCIE-016 (Testing).

0.2 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream chapter and stating only the Memory/Context consequence — never a restatement of the upstream architecture. Major concepts (Memory, Context, Institutional Knowledge, Decision-Time, Knowledge Time, Reconciliation, Knowledge Candidate, Context Envelope, Memory Class) are defined authoritatively once, in the chapter that owns them, and referenced thereafter.

0.3 Governing Invariants Register

These 12 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly:

- ARGUS ≠ INSTITUTIONAL MEMORY.
- MEMORY ≠ CURRENT TRUTH.
- MEMORY ≠ EVIDENCE.
- MEMORY ≠ INSTITUTIONAL KNOWLEDGE.
- INSTITUTIONAL KNOWLEDGE ≠ RULE.
- CURRENT STATE ≠ HISTORICAL STATE ≠ DECISION-TIME STATE.
- SEMANTIC SIMILARITY ≠ AUTHORITY OR PRECEDENT.
- MODEL REPLACEMENT ≠ MEMORY RESET.
- PRIVATE CONTEXT ≠ SHARED CONTEXT.
- CONTEXT PROPAGATION ≠ PERMISSION PROPAGATION.
- REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION.
- PRIOR HUMAN DECISION ≠ CURRENT ACTION AUTHORITY.

0.4 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of six labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SOURCE/WORKFLOW DISCOVERY REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; or SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED — each additionally marked Blocking or Non-Blocking. Chapter 35 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §35.4).

0.5 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual architectural interpretation or Proposed Design Default. Every chapter in this edition is stamped IN DEVELOPMENT — FOR HUMAN REVIEW and remains so until explicitly approved. A technology named from NCIE-004 retains its existing NCIE-004 approval status here — using it in an NCIE-006 example never promotes it.

Contents

This table of contents updates automatically in Microsoft Word. If page numbers do not appear immediately, select the field and choose Update Field.

- 0.1 Authority Chain 2
- 0.2 Token-Conservation Convention 2
- 0.3 Governing Invariants Register 2
- 0.4 Human Review Focus Classification 2
- 0.5 Status Convention 3
- Memory & Context Mandate, Authority Chain & Specification Boundary 8
 - 1.1 Mandate 8
 - 1.2 Specification Boundary — What NCIE-006 Does Not Own 8
 - 1.3 Structural Change Control 8
 - Human Review Focus 8
- Memory Doctrine, Invariants & Epistemic Boundaries 9
 - 2.1 Authoritative Definitions 9
 - 2.2 Epistemic Boundaries..... 9
 - 2.3 Model Independence..... 9
 - Human Review Focus 9
- Memory Class Taxonomy 10
 - 3.1 Memory Class Architecture 10
 - 3.2 Memory Class Matrix 10
 - 3.3 Class-Specific Attributes 11
 - Human Review Focus 11
- Context Taxonomy & Context Envelope 12
 - 4.1 Context Envelope Specification..... 12
 - 4.2 Context Taxonomy..... 12
 - Human Review Focus 12
- Conversation Memory 13
 - 5.1 Conversation Memory Boundary 13
 - 5.2 Lifecycle 13
 - Human Review Focus 13
- User Preference Memory 14
 - 6.1 Preference Memory Boundary 14
- Workspace & Session Continuity Memory..... 15
 - 7.1 Workspace Continuity 15
 - 7.2 Reconciliation on Resume 15
- Collaborative Room Memory..... 16
 - 8.1 Room Memory Is Not a Transcript..... 16
 - 8.2 Voice-Derived Room Memory..... 16
 - 8.3 Room Snapshot and Cross-Session Continuity 16
 - Human Review Focus 16

Investigation Memory 17

 9.1 Investigation Memory Behavior 17

 9.2 Sensitivity Inheritance 17

Situation Memory 18

 10.1 Situation Memory Behavior 18

Decision Memory & Decision-Time Context 19

 11.1 Decision Memory Contents 19

 11.2 Decision-Time Definition..... 19

 11.3 Integrity Protection 19

 11.4 Prior Decision Is Never Automatic Current Authority 19

Historical Memory & Bitemporal Reconstruction 20

 12.1 Bitemporal Definitions 20

 12.2 Historical Reconstruction Flow 20

 12.3 Correction Without Overwrite..... 20

 Human Review Focus 20

Current-State Reconciliation 21

 13.1 Reconciliation States..... 21

 13.2 Current-State Reconciliation Flow..... 21

 13.3 Priority Rule 21

Context Assembly & Minimum-Necessary Retrieval 22

 14.1 Context Assembly Flow..... 22

 14.2 Assembly Considerations 22

 14.3 ARGUS Does Not Access Raw Memory Stores 22

Context Propagation Across Services & Workflows..... 23

 15.1 Context Propagation ≠ Authorization Propagation..... 23

 15.2 Permission-Change Propagation 23

Private, Shared & Institutional Context Boundaries 24

 16.1 Private → Shared → Institutional Context Transition 24

 16.2 Shared Room Disclosure 24

ARGUS Memory Retrieval Interface 25

 17.1 Governed Retrieval Interface 25

 17.2 Retrieval Result Labels 25

Institutional Knowledge Candidate Lifecycle..... 26

 18.1 Knowledge Candidate Definition 26

 18.2 Institutional Knowledge Promotion Lifecycle..... 26

 18.3 Full Lifecycle State Machine..... 26

 18.4 ARGUS Role 26

 Human Review Focus 27

Institutional Knowledge Versioning, Applicability & Review 28

 19.1 Applicability Scope..... 28

 19.2 Review-Due Behavior..... 28

19.3 Versioning 28

Human Review Focus 28

Knowledge Challenge, Contradiction & Impact Analysis 29

20.1 Knowledge Challenge, Supersession & Retraction Lifecycle..... 29

20.2 Retraction Impact Analysis 29

20.3 Re-Review Triggers 29

Knowledge Graph Context & Memory Relationships 30

21.1 Memory Authority Hierarchy 30

21.2 Graph Authorization 30

Dependency Graph & Downstream Impact Tracking..... 31

22.1 Knowledge Graph / Dependency Relationship Architecture 31

22.2 Typed Dependencies and Impact Analysis 31

22.3 Pending-Output Invalidation 31

Memory Search, Retrieval Modes & Ranking..... 32

23.1 Search & Retrieval Architecture 32

23.2 Ranking Factors 32

23.3 Retrieval Explainability..... 32

23.4 Search-Result Leakage Prevention 32

Semantic Memory & Vector Retrieval Boundaries 33

24.1 Embeddings Are Derived, Not Authoritative..... 33

24.2 Sensitive Semantic Indexing..... 33

Memory Security, Authorization & Zero-Trust Controls..... 34

25.1 Memory Authority Matrix 34

25.2 Concurrency and Conflict 34

25.3 Remembered Authorization Is Never Current Authorization..... 34

Sensitive Memory, Protected Identity & Voice Context..... 35

26.1 Governed Sensitive Classes 35

26.2 Voice Object Distinctness 35

Human Review Focus 35

Memory Provenance, Attribution & Audit 36

27.1 Provenance Requirements 36

27.2 No Chain-of-Thought Requirement 36

27.3 AI-Generated Summaries..... 36

27.4 Historical Prompt-Injection Boundary 36

Retention, Legal Hold, Archival & Secure Disposition 37

28.1 Class-Specific Retention 37

28.2 Holds Override Disposition 37

28.3 Derived Deletion..... 37

28.4 Personal Memory Deletion Boundary 37

Human Review Focus 37

Memory Storage Architecture & Canonical/Derived Stores 38

29.1 Logical Memory Storage Architecture 38

29.2 Technology Neutrality 38

Memory APIs, Events & Service Contracts 39

30.1 Service Capabilities 39

30.2 Material Events 39

Scalability, Tiering, Performance & Observability 40

31.1 Tiering and Eventual Consistency 40

31.2 Canonical-State Fallback 40

31.3 Observability — Technical vs Governance Indicators 40

31.4 Performance Boundary 40

Resilience, Backup, Recovery & Post-Restore Reconciliation 41

32.1 Failure and Degraded Modes 41

32.2 Backup / Recovery / Post-Restore Reconciliation Flow 41

32.3 Recovery Priority and Integrity 41

Human Review Focus 42

Legacy Memory Migration & Historical Identity Resolution 43

33.1 Migration Requirements 43

33.2 Technology Migration Neutrality 43

Human Review Focus 43

Memory & Context Testing, Verification & Acceptance 44

34.1 Required Test Categories 44

34.2 Adversarial Memory Tests 44

34.3 Mandatory Acceptance Tests 44

Human Review Focus 45

Human Review Register, Open Decisions & Proposed Defaults 46

35.1 Purpose 46

35.2 Summary 46

35.3 Complete Register 46

35.4 Mechanical Verification 48

35.5 Closure and Blocking Scope 48

Requirements Traceability, Acceptance & Handover Gate 49

36.1 Traceability Matrix 49

36.2 Traceability Mechanical Verification 50

36.3 Acceptance Checklist 50

36.4 Handover 50

Human Review Focus 50

Memory & Context Mandate, Authority Chain & Specification Boundary

Establish NCIE-006 as the detailed Memory/Context authority without redefining upstream architecture.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.2, §20.12 — Memory tiers and the formal promotion boundary are architecturally fixed; NCIE-006 details their implementation. | NCIE-003 Ch.22 — Memory Object, Knowledge Candidate, Institutional Knowledge entities are canonical; NCIE-006 specifies the services and lifecycle around them.

1.1 Mandate

NCIE-006 consolidates persistent Memory and Context architecture into an implementable specification. It preserves model independence, Human-Primary governance, temporal integrity, provenance and authorization, and does not redesign the upstream architecture that already fixes these requirements.

1.2 Specification Boundary — What NCIE-006 Does Not Own

Table 1. NCIE-006 specification boundary.

Concern	Owned By
AI/agent/model orchestration engineering	NCIE-007; NCIE-006 specifies only what Memory/Context must expose to it
Identity, authorization enforcement engineering	NCIE-004 Ch.23; NCIE-009; NCIE-006 specifies authorization requirements, not IAM implementation
Evidence object lifecycle, chain-of-custody engineering	NCIE-003 Ch.17; NCIE-010; NCIE-006 references Evidence, never redefines it
API/event transport engineering	NCIE-004 Ch.8-9; NCIE-011; NCIE-006 specifies service contracts, not wire protocols
Physical database/search/graph/vector technology selection	NCIE-004 (Technology Decision Matrix); NCIE-014; NCIE-006 specifies logical storage roles only
Comprehensive test-suite implementation	NCIE-016; NCIE-006 Ch.34 specifies required test categories, not the full verification programme
ARGUS conversational/analytical behavior	NCIE-005; NCIE-006 specifies what Memory ARGUS may retrieve, not how ARGUS behaves

1.3 Structural Change Control

The 36-chapter structure is locked. A material contradiction or omission discovered during expansion is flagged STRUCTURAL CHANGE REQUIRED — HUMAN REVIEW rather than silently resolved. None has been raised in this edition.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the acceptance authority for NCIE-006 as a whole and for individual chapter sign-off.

Traceability: NCIE-002 Ch.20; NCIE-003 Ch.22.

Memory Doctrine, Invariants & Epistemic Boundaries

Lock Memory ≠ Evidence, Memory ≠ Current Truth, Memory ≠ Institutional Knowledge, Knowledge ≠ Rule, and ARGUS ≠ Memory.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.2 — ARGUS is not institutional memory; Memory ≠ Current Truth is architecturally fixed. | NCIE-003 Ch.22 §22.2 — Institutional Knowledge ≠ Rule; promotion requires recorded human review.

2.1 Authoritative Definitions

Table 2. Core term definitions, stated once and referenced throughout this document.

Term	Definition
Memory	A persistent record of something previously observed, decided, discussed or concluded. It is retrospective by nature — it records what was, not what currently is.
Institutional Knowledge	A Memory-derived generalisation that has been deliberately reviewed and approved through the lifecycle in Ch.18-19, and is applicable within a defined scope (Ch.19 §19).
Evidence	An authoritative artifact with source, acquisition and validation lineage (NCIE-003 Ch.17) — governed entirely outside NCIE-006; Memory may reference Evidence, never substitute for it.
Rule	A governed, effective-dated parameter or threshold (NCIE-003 Ch.24) — governed entirely outside NCIE-006; Institutional Knowledge may motivate a Rule Change Candidate but never directly modifies a Rule.

2.2 Epistemic Boundaries

- ARGUS ≠ Institutional Memory — ARGUS has no memory store of its own; every recollection is retrieved from governed NCIE-006 services (front matter §0.3).
- Memory ≠ Current Truth — a Memory Object is never presented as live authoritative state without reconciliation (Ch.13).
- Memory ≠ Evidence — a Memory Object may reference Evidence; it never becomes Evidence by being remembered.
- Memory ≠ Institutional Knowledge — a raw Memory Object requires deliberate promotion (Ch.18) before it carries institutional authority.
- Institutional Knowledge ≠ Rule — approved Knowledge informs human judgment and may motivate a Rule Change Candidate; it never directly alters a governed Rule (NCIE-003 Ch.24).

2.3 Model Independence

Model Replacement ≠ Memory Reset. Replacing the ARGUS model or provider (NCIE-004 Ch.18) never erases or resets Memory or Institutional Knowledge, since both live in governed NCIE-006 services entirely outside the model/provider boundary. Chapter 34's acceptance tests demonstrate this directly.

Human Review Focus

- **[RESOLVED BY UPSTREAM BASELINE]** All five epistemic boundaries in §2.2 are fixed by NCIE-002 Ch.20 and NCIE-003 Ch.22; no institutional decision is required to adopt them.

Traceability: NCIE-002 Ch.20 §20.2, §20.12; NCIE-003 Ch.22.

Memory Class Taxonomy

Define Conversation, Preference, Workspace, Room, Investigation, Situation, Historical, Decision and Institutional Knowledge memory.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.2 — Private/Shared/Institutional tiers are the architectural basis this taxonomy details into nine operational classes.

3.1 Memory Class Architecture

Figure 1. Memory Class Architecture — authority flows only through deliberate promotion (Ch.18); no Memory class silently escalates its own authority.

SOURCE	RELATIONSHIP	TARGET
Conversation Memory / User Preference Memory	carry no institutional authority into	Workspace, Room, Investigation, Situation Memory
Workspace, Room, Investigation, Situation Memory	may be finalized (never auto-converted) into	Decision Memory
Decision Memory / Historical Memory	may be generalised through deliberate promotion (Ch.18) into	Institutional Knowledge
Institutional Knowledge	never demotes back to, nor is derived automatically from,	Conversation or Preference Memory

3.2 Memory Class Matrix

Nine Memory classes are distinguished throughout this document and never collapsed into one generic store or lifecycle:

Table 3. Memory Class Matrix — nine distinguished classes, each independently governed.

Memory Class	Purpose	Authority	Owner Chapter	Reconciliation Required?
Conversation Memory	Low-authority conversational continuity	None — never overrides current state, Evidence, Rules, Findings, Decisions or authorization	Ch.5	No
User Preference Memory	Low-risk personalization	None — never alters permissions, Evidence meaning, Rules, analytical truth or Decision authority	Ch.6	No
Workspace/Session Continuity Memory	Resumable individual work state	Low — reflects a user's own prior working state	Ch.7	Yes, on resume
Collaborative Room Memory	Structured multi-user analytical reasoning	Working analysis — Hypothesis/Consensus, never Finding/Decision	Ch.8	Yes, on re-entry
Investigation Memory	Investigative evidence relationships, hypotheses, tasks	Working analysis	Ch.9	Yes, on resume
Situation Memory	Evolving cross-domain analytical state	Working analysis	Ch.10	Yes, on resume
Decision Memory	High-integrity Decision-Time snapshots	High — protected historical record of institutional action	Ch.11	N/A — immutable once finalized
Historical Memory	Bitemporal record of past canonical state	Reflects what was known/true at a point in time	Ch.12	Yes, before consequential reuse
Institutional Knowledge	Reviewed, approved, applicability-scoped generalisation	Institutional — requires human approval (Ch.18-19)	Ch.18-19	Yes, via challenge/review (Ch.20)

3.3 Class-Specific Attributes

Each class in §3.2 additionally has its own Scope, Retention, Authorization and Retrieval behavior, detailed in its owning chapter — this matrix is the index, not the full specification for any one class.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm retention, access or ownership rules that differ by Memory class beyond what each owning chapter proposes.

Traceability: NCIE-002 Ch.20 §20.2.

Context Taxonomy & Context Envelope

Define purpose-bounded, authorized, temporal, provenance-aware context passed among users, services, tools and ARGUS.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.19 §19.6 — Conversation/Tool Invocation/Proactive Contribution objects carry resolved context; NCIE-006 formalizes the envelope structure.

4.1 Context Envelope Specification

The Context Envelope is the single implementable structure passed among users, services, tools and ARGUS. Fields are included only where they carry real weight — not merely because they are convenient:

Table 4. Context Envelope field specification.

Field	Purpose
Context ID	Stable reference for this specific context instance (correlation/audit).
User	The human identity the context is resolved for (never a machine-inferred identity, NCIE-005 Ch.4 §4.2).
Session / Workspace / Room / Investigation / Situation / Task	Which of these scopes the context is bound to — mutually clarifying, not all populated simultaneously.
Operator / Domain / Geography	The NCIE-003 canonical scope the context concerns.
Time Mode	Current / Historical / Decision-Time (Ch.12-13) — never left implicit.
Time Range / Decision-Time Cutoff	The specific bound applicable when Time Mode is not Current.
Purpose	Why this context is being assembled (feeds Ch.14’s minimum-necessary retrieval).
Classification	The governing classification tier (NCIE-003 Ch.31) of the context’s most sensitive component.
Authorization Context	The effective authorization the context was assembled under (Ch.25) — never expanded after assembly.
Provenance	Where each component of the context originated (Ch.27).
Correlation ID	Cross-service tracing reference (NCIE-004 Ch.9).

4.2 Context Taxonomy

- Private Context — visible only to the originating user (Ch.16).
- Shared Context — visible to authorized participants of a specific Room/Investigation/Situation (Ch.16).
- Institutional Context — promoted, broadly applicable Institutional Knowledge (Ch.16, Ch.18-19).

Human Review Focus

- **[PROPOSED DESIGN DEFAULT]** The Context Envelope field set in §4.1 is proposed as complete for launch; additional fields may be added additively without a breaking change (NCIE-003 Ch.2 §2.8 pattern).

Traceability: NCIE-002 Ch.19 §19.6.

Conversation Memory

Specify low-authority conversational continuity, retention, summaries and invalidation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.5 — Conversational assistance grounds answers in authoritative tool results; Conversation Memory supplies continuity only, never authority.

5.1 Conversation Memory Boundary

Conversation Memory preserves continuity across turns and sessions. It never overrides Current Authoritative Domain State, Evidence, Rules, Findings, Decisions or authorization. A user previously saying something does not make it current institutional truth — Conversation Memory is retrieved and labelled per NCIE-005 Ch.11's epistemic-type matrix ("Historical Memory"), never presented as a live fact.

5.2 Lifecycle

Table 5. Conversation Memory lifecycle.

State	Trigger	Resulting State
Active	Turn recorded during a live session	Retrievable as recent continuity
Aged	Retention policy threshold reached (Ch.28)	Retrievable as Historical Memory only, not default continuity
Invalidated	User/administrator requests removal, or content found materially inaccurate	Retained per Ch.28 §28 rules (INVALIDATED ≠ DELETED), excluded from active retrieval

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm Conversation Memory retention period — no universal period is assumed; classified per Ch.28's class-specific retention requirement.

Traceability: NCIE-005 Ch.5, Ch.11.

User Preference Memory

Define permitted low-risk preferences without allowing preferences to change authority or truth.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.28 — Personalization is permitted for presentation/notification/voice/verbosity/workspace; preferences never alter authorization or institutional truth.

6.1 Preference Memory Boundary

Table 6. User Preference Memory boundary (mirrors NCIE-005 Ch.28 §28.2, restated here as the canonical Memory-side rule).

Preference May Influence	Preference Shall Never Alter
Presentation, language, voice, notification preferences, accessibility settings	Permissions
Low-risk personalization generally	Evidence meaning
—	Regulatory Rules
—	Analytical truth
—	Decision authority

Traceability: NCIE-005 Ch.28.

Workspace & Session Continuity Memory

Preserve resumable individual work state independently of runtime sessions.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.3 §3.1 — Workspace layout/context state must survive across sessions independently of runtime session objects.

7.1 Workspace Continuity

Workspace Memory preserves an individual's resumable work state — layout, active filters, open panels, last context — independently of any specific runtime session, browser tab or ARGUS conversation instance. A session ending does not destroy the workspace state a user would expect to resume.

7.2 Reconciliation on Resume

On resume, Workspace Memory is reconciled against current authoritative state (Ch.13) before being presented — a saved filter referencing a since-closed Investigation, for example, surfaces as CHANGED or SUPERSEDED rather than silently reopening stale context.

Traceability: NCIE-002 Ch.3 §3.1.

NCIE-006 — Chapter 8 Expansion

Collaborative Room Memory

Persist structured multi-user reasoning independently of the raw transcript.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.5 — Room Memory persists structured reasoning objects, distinct from the raw transcript, per the Collaborative Reasoning State architecture. | NCIE-003 Ch.21 §21.3 — Room, Thread, Question, Hypothesis, Assumption, Contradiction, Evidence Gap, Task, Human Judgment, Decision Point, ARGUS Intervention, Consensus, Dissent and Room Snapshot are canonical persistent objects.

8.1 Room Memory Is Not a Transcript

Room Memory persists the structured analytical objects named above as first-class records — the raw message transcript may supplement this structure but never substitutes for it. A 'summarize this room' request is answered from the structured objects, not by re-reading the transcript from scratch.

8.2 Voice-Derived Room Memory

VOICE-DERIVED MEMORY. Where verbal Collaborative Intelligence is used, Raw Audio, Speech-to-Text Transcript, Structured Room Memory and Formal Record are distinct objects, never automatically the same thing. A transcript error never silently alters a persisted Hypothesis, Human Judgment, Decision Point or authorization. Where a material spoken contribution is mistranscribed, correction preserves the original transcript state where required, the corrected transcript, correcting actor, correction time, and the affected structured objects.

8.3 Room Snapshot and Cross-Session Continuity

A Room Snapshot captures the Collaborative Reasoning State at a point in time, enabling 'where were we?' reconstruction by combining the snapshot with current authoritative state (never presenting a stale snapshot as current without reconciliation, Ch.13).

Human Review Focus

- **[SOURCE/WORKFLOW DISCOVERY REQUIRED]** Confirm audio retention policy specifics with NCA (inherits NCIE-005's open item on audio/transcript retention as a separate decision, per §0.2's no-duplication rule — see NCIE-005 Ch.29 §29 review).

Traceability: NCIE-002 Ch.20 §20.5, §20.10; NCIE-003 Ch.21.

Investigation Memory

Preserve investigative evidence relationships, hypotheses, contradictions, gaps, tasks and human judgments.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.15 §15.3 — Investigation stage progression (Detection→Suspicion→Investigation→Validation→Blocking→Outcome) is canonical; Investigation Memory persists its working analytical state.

9.1 Investigation Memory Behavior

Investigation Memory preserves evidence relationships, hypotheses, contradictions, evidence gaps, tasks and human judgments accumulated during an investigation, supporting pause/resume across sessions, shifts, users, ARGUS model changes and system restarts — resumption never depends on the original chat/session being available.

9.2 Sensitivity Inheritance

Investigation Memory tied to Anti-Fraud/SIMBOX work inherits sensitive-tier classification (NCIE-003 Ch.15, Ch.31) automatically; it is never treated as ordinary working-analysis memory merely because it uses the same underlying structure as Ch.8's Room Memory.

Traceability: NCIE-003 Ch.15.

Situation Memory

Preserve evolving Situation context, timelines and cross-domain analytical state.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.18 — Situation is a canonical entity distinct from Condition/Alert/Investigation/Case; Situation Memory persists its evolving cross-domain analytical state.

10.1 Situation Memory Behavior

Situation Memory preserves timelines and cross-domain analytical state across the Situation's lifecycle (NCIE-003 Ch.18), supporting the same pause/resume guarantees as Investigation Memory (Ch.9) — across sessions, shifts, users, model changes and restarts.

Traceability: NCIE-003 Ch. 18.

Decision Memory & Decision-Time Context

Preserve high-integrity snapshots of what was known when consequential human Decisions were made.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.16 §16.3, Ch.25 §25.5 — Decision-Time Snapshot and Decision-Time Context are canonical; Decision Memory is their high-integrity persisted form.

11.1 Decision Memory Contents

Decision Memory preserves, where applicable:

- Decision ID, Decision Authority, Decision Time.
- Decision-Time Snapshot (the bounded fact/evidence/rule-version set available at that timestamp).
- Evidence Available at Decision Time; Applicable Rule Versions.
- Known Hypotheses, Known Contradictions, Known Evidence Gaps at that time.
- Human Rationale; ARGUS Contribution (distinctly attributed, NCIE-005 Ch.27).
- Subsequent Review Relationship (appeal/reversal, linked forward, never rewriting the original).

11.2 Decision-Time Definition

Decision-Time is formally defined here and referenced, not redefined, elsewhere in this document: the specific timestamp as of which a Decision's supporting context is bound. Reconstruction implements AS KNOWN AT T, never WHAT WE KNOW NOW ABOUT T — the system supports both views but never confuses them (Ch.12 covers the general bitemporal mechanism this relies on).

11.3 Integrity Protection

Table 7. Decision Memory integrity requirements.

Requirement	Behavior
Immutability	A finalized Decision-Time Snapshot is protected against ordinary modification once created
Subsequent events as new objects	Correction/review/appeal/outcome are linked as subsequent objects/events, never rewrites of the original snapshot
Sensitive-content access	Access to a Decision record does not automatically grant access to every underlying restricted Evidence item (Ch.25); redacted/limited reconstruction is supported
Outcome ≠ quality/causation	A later-known outcome informs learning; it never automatically rewrites the historical assessment of whether the Decision was reasonable given what was known then

11.4 Prior Decision Is Never Automatic Current Authority

PRIOR HUMAN DECISION ≠ CURRENT ACTION AUTHORITY (front matter §0.3). A finalized Decision recorded in Decision Memory may inform current human reasoning — as precedent, context or pattern — but it is never automatically replayed as authorization for a new blocking, tracking, regulatory or other consequential action. A new consequential action requires its own current human Decision Authority and current Decision-Time context (§11.1-§11.2); ARGUS may surface a prior Decision as relevant context (NCIE-005 Ch.24) but never initiates or authorizes a new action on the strength of the prior Decision alone.

Traceability: NCIE-003 Ch. 16, Ch.25, Ch.26.

Historical Memory & Bitemporal Reconstruction

Implement occurrence/effective time and knowledge/record time for defensible historical reconstruction.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.5 — Occurrence/effective time and knowledge/record time are the canonical bitemporal axes; this chapter is their Memory-service realisation.

12.1 Bitemporal Definitions

Table 8. Bitemporal time axes (formally defined here; NCIE-003 Ch.5 establishes the platform-wide principle this chapter applies to Memory specifically).

Time Axis	Definition
Occurrence / Valid Time	When the remembered fact actually happened or was true.
Knowledge / Record Time	When NCIE came to know/record the fact — never substituted with ingestion time where they genuinely differ.

12.2 Historical Reconstruction Flow

Figure 2. Decision-Time / historical reconstruction flow — later evidence never contaminates the reconstructed view.

01	Reconstruction request: 'what was known as of knowledge-time T?'
02	Bind to canonical facts with knowledge time ≤ T (NCIE-003 Ch.25 pattern)
03	Bind to Rule Versions effective at T (NCIE-003 Ch.24)
04	Bind to Evidence registered with knowledge time ≤ T
05	Exclude any fact/rule/evidence with knowledge time > T
06	Return AS KNOWN AT T, with later developments available separately as SUBSEQUENTLY LEARNED

12.3 Correction Without Overwrite

Historical knowledge state is never overwritten merely because current understanding changes. A correction to historical Memory uses an explicit CORRECTION, AMENDMENT, NEW VERSION, INVALIDATION or SUPERSESSSION relationship (Ch.20 detail for Knowledge specifically) — the institution's prior understanding remains reconstructable even after it is known to have been wrong.

Human Review Focus

- [PROPOSED DESIGN DEFAULT]** UTC as the canonical knowledge-time storage basis is proposed, consistent with NCIE-003 Ch.5 §5.8's platform-wide default.

Traceability: NCIE-003 Ch.5, Ch.25.

Current-State Reconciliation

Reconcile remembered state against authoritative current domain state before consequential reuse.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.10 — Previous collaborative state must be reconciled with current authoritative NCIE state — the architectural mandate this chapter operationalizes for all Memory classes.

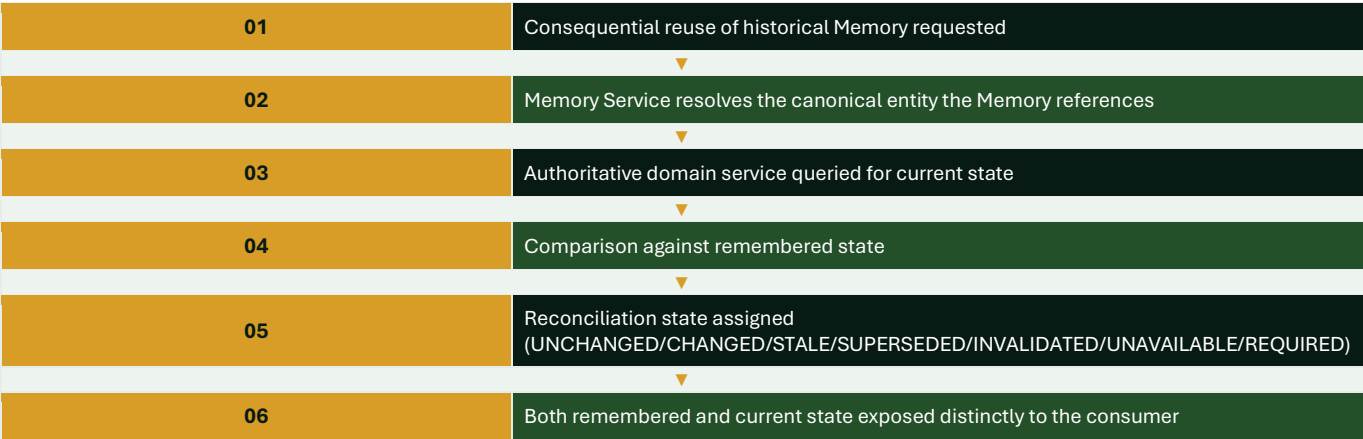
13.1 Reconciliation States

Table 9. Current-State Reconciliation states — historical Memory is never silently updated; history and current state are always separately exposed.

State	Meaning
UNCHANGED	Current authoritative state matches remembered state exactly.
CHANGED	Current state differs from remembered state in a way that does not invalidate the memory's historical accuracy.
STALE	Remembered state has exceeded its freshness expectation and has not been re-verified.
SUPERSEDED	A newer canonical version exists; the remembered version is no longer current.
INVALIDATED	The remembered fact has been formally invalidated at its source.
CURRENT STATE UNAVAILABLE	The authoritative source cannot be reached to verify (Ch.30 §30 failure behavior).
RECONCILIATION REQUIRED	A material discrepancy exists that requires explicit resolution before consequential reuse.

13.2 Current-State Reconciliation Flow

Figure 3. Current-State Reconciliation flow.



13.3 Priority Rule

When a user requests current operational state, authoritative domain services take precedence over remembered historical state. Memory may provide context; it never replaces authoritative current-state retrieval (NCIE-005 Ch.23 restates this at the ARGUS behavior layer; this chapter is its Memory-service-level enforcement point).

Traceability: NCIE-002 Ch.20 §20.10; NCIE-005 Ch.23.

Context Assembly & Minimum-Necessary Retrieval

Assemble only authorized, relevant and purpose-bounded context for humans, ARGUS and tools.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.24 §24.3 — Context Builder assembles only authorization-filtered, minimal context per request — this chapter is its Memory-side implementation.

14.1 Context Assembly Flow

Figure 4. Context Assembly Flow — minimum-necessary retrieval, never a full institutional context dump.

01	Requesting service/ARGUS states Purpose and Context Envelope (Ch.4)
02	Authorization filter applied (Ch.25) — never expanded beyond requester's effective scope
03	Classification filter applied — sensitive-tier content requires explicit eligibility
04	Temporal Mode applied (Current/Historical/Decision-Time, Ch.11-13)
05	Relevance and Applicability filters applied (Ch.19 §19 scope matching for Institutional Knowledge)
06	Context Budget enforced — excess content trimmed by materiality, not truncated arbitrarily
07	Assembled Context Envelope returned

14.2 Assembly Considerations

- Purpose — what the context is for shapes what is retrieved.
- Authorization and Classification — never assembled beyond the requester's effective scope.
- Temporal Mode — Current/Historical/Decision-Time are never mixed silently.
- Relevance and Applicability — Institutional Knowledge outside its applicable scope (Ch.19) is excluded, not force-included.
- Data Minimization and Context Budget — prioritised by authorization, purpose, authority, temporal applicability, materiality, evidence/contradiction relevance; never by semantic similarity alone.

14.3 ARGUS Does Not Access Raw Memory Stores

Figure 5. ARGUS's memory access path — no AI-only unrestricted retrieval path exists.

01	ARGUS
02	Context/Memory Retrieval Service
03	Authorization + Purpose + Temporal Filters
04	Memory/Knowledge Sources

Traceability: NCIE-002 Ch.24 §24.3.

Context Propagation Across Services & Workflows

Propagate canonical context without propagating permissions or corrupting temporal meaning.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.4 §4.4 — Zero-trust authorization is re-evaluated at every service boundary; context propagation never substitutes for it.

15.1 Context Propagation ≠ Authorization Propagation

A user navigating Dashboard → Situation → Investigation → Evidence → ARGUS may retain relevant context (operator, geography, active thread). Navigation never expands access — every destination independently applies its own authorization check (Ch.25), regardless of what context was carried forward.

15.2 Permission-Change Propagation

- If a user's authorization changes during an active session, Room or Investigation, subsequently assembled context reflects the new authorization immediately — authorization is never assumed valid for the lifetime of a session.
- Revocation after sensitive context has already been retrieved is handled via: future retrieval block; UI redaction/removal where technically feasible; cache invalidation; ARGUS context removal; search/graph access restriction; and audit. Already-perceived human information cannot be technically "unseen" — this is an acknowledged limit, not a false guarantee.

Traceability: NCIE-002 Ch.4 §4.4.

Private, Shared & Institutional Context Boundaries

Govern movement from private context to Room-shared context and ultimately to reviewed institutional knowledge.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.9 — Shared room context never pools participant permissions; Private User Memory is not automatically shareable.

16.1 Private → Shared → Institutional Context Transition

Figure 6. Private → Shared → Institutional context transition — each step is a deliberate, human-governed action, never automatic.

01	Private Context (single user, e.g. Preference/Conversation Memory, Ch.5-6)
02	Deliberate sharing action into a Room/Investigation/Situation (Ch.8-10)
03	Shared Context — visible only to that Room/Investigation/Situation's authorized participants
04	Knowledge Candidate proposed from Shared Context pattern (Ch.18)
05	Human review and approval (Ch.18-19)
06	Institutional Context — Approved Institutional Knowledge, broadly applicable within its scope

16.2 Shared Room Disclosure

A Collaborative Room's shared disclosure is calculated per-participant from governed authorization, never as a permission union: one participant's ability to access an object never means ARGUS may expose it to every Room participant (NCIE-002 Ch.20 §20.9). ARGUS's effective Room context is restricted to information appropriately shareable in that context — private context from one participant never silently becomes ARGUS-shared Room context.

Traceability: NCIE-002 Ch.20 §20.9; NCIE-003 Ch.4 §4.72-4.79.

ARGUS Memory Retrieval Interface

Define governed Memory Service interfaces consumed by ARGUS without giving ARGUS unrestricted institutional memory access.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.23 — ARGUS retrieves memory via governed services and reconciles it against current state; NCIE-006 specifies the interface those retrievals actually use.

17.1 Governed Retrieval Interface

ARGUS consumes Memory exclusively through the Memory/Context Retrieval Service (Ch.14's assembly flow), never a direct store connection. Every retrieval carries the requesting identity's effective authorization, purpose, and temporal mode; none of these may be widened by the fact that the requester is ARGUS rather than a human.

17.2 Retrieval Result Labels

Returned Memory/Knowledge visibly distinguishes its category — never collapsed into one generic "memory result":

- CURRENT AUTHORITATIVE CONTEXT
- HISTORICAL MEMORY
- DECISION-TIME CONTEXT
- APPROVED INSTITUTIONAL KNOWLEDGE
- PRIOR HUMAN JUDGMENT
- ARGUS-GENERATED HISTORICAL ANALYSIS
- UNVALIDATED KNOWLEDGE CANDIDATE

These labels are the Memory-service-side source of NCIE-005 Ch.11's ARGUS Output/Epistemic-Type Matrix — that chapter specifies how ARGUS presents them; this chapter specifies that the Memory Service produces them accurately in the first place.

Traceability: NCIE-005 Ch. 11, Ch.23.

Institutional Knowledge Candidate Lifecycle

Define how lessons and reusable insights become governed Knowledge Candidates.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.22 §22.4 — Promotion to Institutional Knowledge requires a recorded human Review State; a Knowledge Candidate never self-promotes.

18.1 Knowledge Candidate Definition

A Knowledge Candidate is a proposed generalisation derived from Memory, awaiting human review before it may carry institutional authority (Ch.2 §2.1's authoritative definition applies).

18.2 Institutional Knowledge Promotion Lifecycle

Figure 7. Institutional Knowledge Promotion Lifecycle — the main forward path; §18.3's table below covers the full state machine including challenge/supersession/retraction branches.

01	CANDIDATE (proposed, origin traceable)
02	REVIEW (human evaluation, separation of duties where policy requires)
03	APPROVED (applicability scope defined, Ch.19)
04	ACTIVE (in force within scope)

18.3 Full Lifecycle State Machine

Table 10. Institutional Knowledge Candidate lifecycle state machine.

State	Trigger	Permitted Actor	Resulting State
CANDIDATE	Proposed from Human Analyst, Collaborative Room, Investigation, Situation, Post-Incident Review, Expert Contribution or ARGUS Proposal	Any of the above; origin remains traceable after promotion	REVIEW
REVIEW	Submitted for human evaluation	Authorized reviewer, distinct from proposer where policy requires (Ch.20 §20.5)	APPROVED or REJECTED
APPROVED	Reviewer approves with defined applicability scope (Ch.19)	Authorized approver	ACTIVE
ACTIVE	In force within its applicability scope	N/A (state, not action)	CHALLENGED, REVIEW_DUE, SUPERSEDED, RETRACTED or EXPIRED
CHALLENGED	Material conflict identified (Ch.20)	Any authorized user/ARGUS may flag; only a human resolves	ACTIVE (challenge resolved) or REVIEW (re-review triggered)
REVIEW_DUE / SUPERSEDED / RETRACTED / EXPIRED	Review date reached / newer version approved / formally withdrawn / applicability period lapsed	Per Ch.19 policy	Terminal or REVIEW, per Ch.19 §19

18.4 ARGUS Role

ARGUS may propose a Knowledge Candidate, organize supporting evidence, surface contradictions, and prepare review material. ARGUS does not approve — this remains true even where the candidate originated from ARGUS itself (NCIE-005 Ch.24 §24.1 restates this at the behavioral layer).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm whether separation of duties between Candidate Contributor, Reviewer and Approver is institutionally required, or whether one authorized role may perform more than one step.

Traceability: NCIE-003 Ch.22 §22.4; NCIE-005 Ch.24.

Institutional Knowledge Versioning, Applicability & Review

Define scope, effective period, limitations, review, challenge, supersession and retraction.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.22 §22.3 — Applicability Scope and Knowledge Version are canonical Institutional Knowledge attributes.

19.1 Applicability Scope

Approved Institutional Knowledge identifies its applicable Domain, Operator, Geography, Technology, Time, Methodology, Situation Type, Regulatory Context or other relevant boundary. Contextual lessons are never silently universalized — if retrieved Knowledge appears used outside its approved scope, ARGUS may warn KNOWLEDGE APPLICABILITY REVIEW REQUIRED (NCIE-005 Ch.24 restates this as ARGUS behavior); it never silently extends the scope itself.

19.2 Review-Due Behavior

Table 11. Institutional Knowledge review and ownership-transfer behavior.

Situation	Behavior
Review date reached	Knowledge class/policy determines whether it remains ACTIVE — REVIEW DUE or moves to SUSPENDED PENDING REVIEW; where policy is unknown, classified INSTITUTIONAL CONFIRMATION REQUIRED, never assumed universally
Ownership changes (staff departure, reorganization, role/department change)	Original Authorship, Review, Approval and Contribution Provenance are preserved; changing the owner never rewrites history
No accountable current owner	Enters GOVERNANCE REVIEW REQUIRED rather than remaining indefinitely ownerless

19.3 Versioning

Each Knowledge Version retains its own effective period and content; retrieval of current guidance always resolves to the active version, while historical reconstruction (Ch.12) can recover which version was in force at a prior knowledge time.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm review-due default behavior (ACTIVE-REVIEW-DUE vs SUSPENDED) per Knowledge class where NCA policy is not otherwise specified.

Traceability: NCIE-003 Ch.22 §22.3.

Knowledge Challenge, Contradiction & Impact Analysis

Handle conflicts between current evidence and approved knowledge through explicit review and downstream impact analysis.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.24 — ARGUS may identify a contradiction between current Evidence and approved Knowledge; humans govern the Knowledge response.

20.1 Knowledge Challenge, Supersession & Retraction Lifecycle

Figure 8. Knowledge Challenge / Supersession / Retraction lifecycle.

01	Material conflict identified between current Evidence and Approved Knowledge (human or ARGUS)
02	KNOWLEDGE REVIEW REQUIRED state generated — never silently resolved by picking one side
03	Human review of the conflict
04	Outcome: Knowledge reaffirmed, superseded by a new version, or retracted
05	SUPERSEDED/RETRACTED recorded — historical traceability retained (SUPERSEDED ≠ DELETED; RETRACTED ≠ NEVER EXISTED)
06	Current retrieval stops presenting superseded/retracted Knowledge as active guidance

20.2 Retraction Impact Analysis

If Knowledge is retracted, active Rooms, Investigations, Situations, Briefings, Reports or Decision-Support Packages that materially relied upon it are flagged for review — historical artifacts are never silently rewritten as a result (Ch.22 formalizes the dependency-tracking mechanism this relies on).

20.3 Re-Review Triggers

Review date reached, material contradiction, underlying Rule change, definition change, methodology change, major incident, repeated challenge, or scope-expansion request may each trigger review — none of them automatically retracts Knowledge; they trigger the human review process in Ch.18 §18.2.

Traceability: NCIE-005 Ch.24; NCIE-003 Ch.22.

Knowledge Graph Context & Memory Relationships

Use canonical graph relationships for memory/knowledge without converting connectivity into fact, cause or culpability.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.19 §19.2 — Graph Relationship ≠ Causality is the platform-wide invariant; this chapter applies it to Memory/Knowledge relationships specifically.

21.1 Memory Authority Hierarchy

Figure 9. Memory Authority Hierarchy — the chain of deliberate promotion from Evidence to Institutional Knowledge.

SOURCE	RELATIONSHIP	TARGET
Evidence (NCIE-003 Ch.17)	may be cited by	Memory Object
Memory Object	may be generalised into	Knowledge Candidate (Ch.18)
Knowledge Candidate	requires human Review State to become	Approved Institutional Knowledge
Approved Institutional Knowledge	may motivate (never directly modify)	Rule Change Candidate (NCIE-003 Ch.24)
Memory / Knowledge Graph edges	are correlations, never	Causality or Culpability claims (NCIE-003 Ch.19 §19.2)

21.2 Graph Authorization

Graph traversal authorizes Node, Edge, Connected Object and, where relevant, Topology Disclosure. A hidden node must not be inferable merely through degree count, neighbor count, path existence or graph visualization — the same authorization discipline NCIE-003 Ch.19 §19.7 requires for the Intelligence Graph applies identically to the Memory/Knowledge Graph.

Traceability: NCIE-003 Ch.19 §19.2, §19.7.

Dependency Graph & Downstream Impact Tracking

Track typed dependencies so corrections and invalidations expose affected downstream artifacts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.25 §25.4 — Correction Relationships trigger forward impact analysis via Dependency Edges; this chapter applies that mechanism to Memory/Knowledge.

22.1 Knowledge Graph / Dependency Relationship Architecture

Figure 10. Knowledge Graph / Dependency Relationship Architecture — typed dependencies used for impact tracking, not causal assertion (Ch.21 §21.1 restates the authority-hierarchy view of this same graph).

SOURCE	RELATIONSHIP	TARGET
Source	supports	Evidence
Evidence	supports	Memory / Hypothesis
Hypothesis	may inform (never automatically becomes)	Finding
Finding	supports	Decision
Decision / Memory	may generalise to	Institutional Knowledge
Institutional Knowledge / Decision	may be cited by	Report / Briefing

22.2 Typed Dependencies and Impact Analysis

If upstream Evidence changes, dependent objects identified via §22.1's typed edges are flagged for review — this never automatically reverses a human Decision (NCIE-003 Ch.25 §25.4's Impact-Review State pattern applies identically here).

22.3 Pending-Output Invalidation

If ARGUS has generated a draft briefing or analytical artifact from information subsequently invalidated before human approval, the artifact is flagged SOURCE STATE CHANGED — REVIEW REQUIRED rather than silently remaining apparently current.

Traceability: NCIE-003 Ch.25 §25.4.

Memory Search, Retrieval Modes & Ranking

Define Current, Historical and Decision-Time search with authority- and applicability-aware ranking.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.4 §4.89-4.93 — Search enforces authorization before returning results, including facets, snippets, autocomplete and semantic matches.

23.1 Search & Retrieval Architecture

Figure 11. Memory Search & Retrieval Architecture — mode-differentiated, authorization-filtered throughout.

01	Search Request (Current / Historical / Decision-Time mode)
02	Authorization Filter
03	Mode-Specific Query (canonical store / bitemporal store / decision-time-bound store)
04	Ranking (Ch.23 §23.2)
05	Labelled, Authorized Results (Ch.17 §17.2 result labels)

23.2 Ranking Factors

Ranking considers Authority, Applicability, Temporal Validity, Governance State, Relevance and Source Quality — never solely Recency or Semantic Similarity, which are inputs to relevance but never the entire ranking basis.

23.3 Retrieval Explainability

For material retrieval, authorized users can determine why an object was returned, along the dimensions in §23.2 plus Source and Retrieval Mode — without exposing implementation-sensitive scoring internals unnecessarily.

23.4 Search-Result Leakage Prevention

Authorization applies before disclosure through titles, snippets, counts, graph neighbors, semantic matches, autocomplete or facets — none of these surfaces is exempt from the same authorization check applied to the underlying object (NCIE-003 Ch.4 §4.89-4.93).

Traceability: NCIE-003 Ch.4 §4.89-4.93.

Semantic Memory & Vector Retrieval Boundaries

Use embeddings for discovery without making vector representations Evidence or Institutional Knowledge.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.26 §26.3 — Sensitive semantic/vector indexing requires isolation; NCIE-004 Ch.16 defines the classification-aware vector retrieval stack this chapter governs the Memory-side rules for.

24.1 Embeddings Are Derived, Not Authoritative

Embeddings are derived retrieval artifacts. They are not Evidence, Institutional Knowledge or canonical truth. Changing the embedding model never changes the substantive meaning or status of the underlying object — a re-embedding event (NCIE-004 Ch.16) is a retrieval-technology change, not a Memory/Knowledge change.

24.2 Sensitive Semantic Indexing

Highly sensitive information — Protected Identity, raw Ghana Card/Passport values, precise Anti-Fraud location, restricted Investigation Evidence, sensitive infrastructure data — is never placed into general-purpose vector infrastructure merely because semantic retrieval is useful. Where semantic retrieval over such content is required, it uses isolated, authorization-aware infrastructure per NCIE-004 Ch.16's index-isolation-by-classification-tier requirement.

Traceability: NCIE-002 Ch.26 §26.3; NCIE-004 Ch.16.

Memory Security, Authorization & Zero-Trust Controls

Apply object-, purpose-, classification- and temporal-aware authorization to all memory classes.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.4 — Zero-trust, four-layer authorization (role/object/field/action) applies to every Memory class independently.

25.1 Memory Authority Matrix

Table 12. Memory Authority Matrix — object-, purpose-, classification- and temporal-aware; nothing is transitively assumed.

Access Granted	Does NOT Automatically Grant
Access to current canonical objects	Historical access, Private Memory access, Room Memory access, Investigation Memory access, or Decision Memory access
Room membership	Access to every Evidence object referenced within the Room (NCIE-003 Ch.4 §4.69)
Access to a Decision record	Access to every underlying restricted Evidence item cited by it (Ch.11 §11.3)

25.2 Concurrency and Conflict

Concurrent human/system activity (two reviewers editing a Knowledge Candidate, a Room Snapshot during active collaboration, retraction while ARGUS retrieves the same object, permission revocation mid-session, correction during historical reconstruction) is never resolved by silent last-write-wins for consequential institutional state. A VERSION CONFLICT state requires reload, merge or human resolution according to object type.

25.3 Remembered Authorization Is Never Current Authorization

REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION (front matter §0.3). A Memory object may record that a person held a role, permission or access grant at some past point — including within a Decision-Time Snapshot (Ch.11) or a Historical Memory reconstruction (Ch.12). That recorded fact is never itself treated as authorizing a new consequential action today. Every consequential action is authorized from the current governed IAM/authorization context (NCIE-004 Ch.23; NCIE-009), re-checked at the time of the new action, regardless of what Memory shows the person was permitted to do previously — a revoked, expired or role-changed permission is never reinstated by historical recollection. Current-State Reconciliation (Ch.13) applies to remembered authorization exactly as it applies to any other remembered state.

Traceability: NCIE-002 Ch.4.

NCIE-006 — Chapter 26 Expansion

Sensitive Memory, Protected Identity & Voice Context

Govern Protected Identity, Anti-Fraud location, voice transcripts/audio and other heightened-sensitivity context.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.14 §14.6, Ch.36 §36.3 — Ghana Card/Passport and Anti-Fraud location protections apply to Memory exactly as to canonical data. | NCIE-005 Ch.29 §29.1 — The Voice Interaction & Spoken-Disclosure Matrix governs what may be spoken; this chapter governs what is remembered about it.

26.1 Governed Sensitive Classes

- Protected Identity (Ghana Card, Passport) — masked/tokenized in Memory exactly as in canonical stores (NCIE-003 Ch.14 §14.6); never stored in plaintext in a Memory Object merely because it appeared in a conversation.
- Anti-Fraud location — inherits sensitive-tier isolation and heightened authorization wherever it appears in Memory (Ch.9 §9.2).
- Restricted Investigation Evidence and sensitive infrastructure data — classification inherited from the referenced canonical object, never downgraded by being remembered.

VOICE-DERIVED MEMORY. Voice/Speaker Recognition ≠ Authenticated Identity — Memory never records an inferred speaker identity as authoritative; identity comes from the governed NCIE authentication context (NCIE-005 Ch.4 §4.2), never from voice characteristics. Display Authorization ≠ Spoken-Disclosure Authorization — remembered authorization to view sensitive information never becomes permission to speak it aloud later; every synthesis still passes through NCIE-005 Ch.29's spoken-disclosure check at the time of speaking. Spoken-notification Memory: the fact that ARGUS spoke a notification is never treated as proof a human heard, understood, acknowledged or acted on it — acknowledgement, where it matters, is a separately stored governed event (Ch.11-style integrity, not inferred from audio).

26.2 Voice Object Distinctness

Raw Audio, Speech-to-Text Transcript, Structured Room/Investigation Memory and Formal Record are distinct objects (Ch.8 §8.2 for the Room-specific case). Audio retention and transcript retention are separate decisions — raw audio storage is never assumed merely because transcription occurred.

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm audio/transcript retention periods and applicable privacy policy — this is the same open question already raised at NCIE-005 Ch.29 §29-review; not duplicated here as a new decision.
- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED]** Confirm cross-border processing boundaries for any voice/transcript content that could leave an approved jurisdiction via cloud STT/TTS processing (NCIE-004 Ch.18's provider eligibility applies).

Traceability: NCIE-003 Ch.14 §14.6, Ch.36; NCIE-005 Ch.29.

Memory Provenance, Attribution & Audit

Preserve human, ARGUS, workflow and source origins plus material retrieval and lifecycle audit.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.27 — AI origin is preserved permanently; human modification never erases it; no private chain-of-thought is stored.

27.1 Provenance Requirements

Material Memory/Knowledge objects preserve:

- Origin and Actor Type (human, ARGUS, workflow, source — always distinguishable).
- Source References; Created Time; Knowledge Time.
- Version; Classification; Review State.
- Correction/Supersession Relationships.

27.2 No Chain-of-Thought Requirement

Memory is never designed to store private model chain-of-thought. Defensibility comes from persisted structured artifacts — Questions, Hypotheses, Assumptions, Contradictions, Evidence Gaps, Evidence References, Tool Results, ARGUS Contributions and Human Responses — matching NCIE-005 Ch.27 §27.2 exactly, restated here as a Memory-architecture constraint rather than an ARGUS behavioral one.

27.3 AI-Generated Summaries

An ARGUS-generated Memory summary retains ARGUS Origin, Source References, Generation Time, applicable temporal cutoff, and Validation/Review State where required. Human adoption never erases AI provenance. A dependent summary whose underlying Memory materially changes supports STALE/REFRESH_REQUIRED rather than remaining silently presented as current.

27.4 Historical Prompt-Injection Boundary

Stored Conversation, Document, Evidence, Knowledge Candidate or Historical Note content may contain malicious instructions. Such content remains data — it never becomes executable ARGUS instruction merely because it is retrieved from Memory (NCIE-002 Ch.24 §24.4's ingestion principle applies identically to retrieval from Memory).

Traceability: NCIE-002 Ch.24 §24.4; NCIE-005 Ch.27.

Retention, Legal Hold, Archival & Secure Disposition

Define differentiated retention, hold, archive and deletion behavior including derived artifacts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.31 — Classification/privacy/retention taxonomy is canonical; this chapter applies class-specific retention to each of Chapter 3's nine Memory classes.

28.1 Class-Specific Retention

No universal Memory retention period exists. Each Memory class (Ch.3 §3.2) has its own retention policy hook; where the institutional/legal period is unknown, it is classified LEGAL/POLICY CONFIRMATION REQUIRED, never invented.

28.2 Holds Override Disposition

LEGAL HOLD, REGULATORY HOLD, INVESTIGATION HOLD and RECORDS HOLD override ordinary disposition for any Memory object they apply to, exactly as NCIE-003 Ch.31 §31.5 requires for canonical data generally.

28.3 Derived Deletion

Where an object is lawfully destroyed, its Search Index entry, Vector Embedding, Cache entry, Graph Materialization and Snippet are addressed explicitly — primary data is never deleted while retrievable derived copies remain unintentionally (NCIE-003 Ch.31 §31.4's Deletion Propagation mechanism applies to every derived Memory representation).

28.4 Personal Memory Deletion Boundary

Deletion of eligible personal Conversation or Preference Memory never silently destroys independent Formal Records, Decision Evidence, Room Records, Investigation Records or Audit Records that have a separate lawful institutional basis.

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm retention periods per Memory class (Ch.3 §3.2) with NCA Legal/Records Management; none is assumed.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm how personal deletion rights interact with institutional records obligations where they may conflict — no universal precedence is assumed.

Traceability: NCIE-003 Ch.31.

Memory Storage Architecture & Canonical/Derived Stores

Define transactional, object, analytical, graph, search/vector and immutable storage roles without creating alternate truth.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.11-17 — Transactional/analytical/object/search/graph/vector/cache store roles are engineering-governed; this chapter assigns Memory's logical role to each without naming a product.

29.1 Logical Memory Storage Architecture

Figure 12. Logical Memory Storage Architecture — canonical vs derived roles; no alternate truth exists in a derived store.

SOURCE	RELATIONSHIP	TARGET
Transactional Memory Metadata	canonical for	Memory Object identity, state, provenance
Object/Document Store	canonical for	Room transcripts, briefing artifacts, large content bodies
Historical/Analytical Store	derived, rebuildable, canonical for	bitemporal historical query (Ch.12)
Knowledge Graph	derived, canonical for	typed Memory/Knowledge relationships (Ch.21)
Search Index / Vector Index	derived, rebuildable, never independently canonical	discovery and semantic retrieval (Ch.23-24)
Snapshot/Immutable Archive	canonical, append-only for	Decision-Time Snapshots and Room Snapshots (Ch.11, Ch.8)
Cache	derived, ephemeral accelerator over	all of the above

29.2 Technology Neutrality

This chapter specifies logical roles only. Physical technology selection remains governed by NCIE-004's Technology Decision Matrix; a technology named there retains its existing approval status when referenced here — this document never promotes a Proposed Design Default to Approved merely by using it in an example (NCIE-004 Ch.33 §33.1A's invariant applies identically).

Traceability: NCIE-004 Ch.11-17, Ch.33.

Memory APIs, Events & Service Contracts

Define stable retrieval, reconciliation, historical-state and knowledge-lifecycle interfaces and events.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.8-9 — API/event contract engineering is governed upstream; this chapter defines the logical capability set those contracts must expose.

30.1 Service Capabilities

- GetCurrentContext, GetHistoricalContext, GetDecisionTimeContext — the three retrieval modes (Ch.23) as distinct calls, never one undifferentiated "get memory" endpoint.
- SearchMemory — authorization- and mode-aware (Ch.23).
- ReconcileMemory — invokes Ch.13's reconciliation flow explicitly.
- CreateKnowledgeCandidate, ReviewKnowledgeCandidate, ChallengeKnowledge, SupersedeKnowledge, RetractKnowledge — one call per lifecycle transition (Ch.18, Ch.20), never one generic "update knowledge" call.

Names may be normalized during downstream implementation (NCIE-011); the capability boundaries above are what must not be collapsed.

30.2 Material Events

- MemoryCreated, MemoryCorrected, MemoryInvalidated.
- KnowledgeCandidateCreated, KnowledgeApproved, KnowledgeChallenged, KnowledgeSuperseded, KnowledgeRetracted.
- DecisionSnapshotCreated.
- ReconciliationRequired.

Event payloads reference sensitive objects rather than copying them unnecessarily (NCIE-003 Ch.28 §28.2's data-minimization pattern applies identically to Memory events).

Traceability: NCIE-004 Ch.8-9; NCIE-003 Ch.28.

Scalability, Tiering, Performance & Observability

Define Hot/Warm/Cold tiering, retrieval SLOs, index health, reconciliation monitoring and governance metrics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.31, Ch.35 — Hot/Warm/Cold tiering and performance targets are NCIE-004's proposed defaults; this chapter applies them to Memory specifically.

31.1 Tiering and Eventual Consistency

Derived representations (Search, Graph, Vector, Cache, Analytical) may update asynchronously. Each derived representation exposes whether it is CURRENT, LAGGING, STALE or UNAVAILABLE — a consequential current-state decision never depends unknowingly on stale derived state (Ch.13's reconciliation states are the canonical-store analogue of this derived-store concept).

31.2 Canonical-State Fallback

Where a derived store is stale or unavailable, the operation may fall back to the canonical store where technically feasible. Where fallback is not feasible, the limitation is exposed explicitly — never fabricated as a complete result.

31.3 Observability — Technical vs Governance Indicators

Table 13. Memory observability — technical and governance indicators tracked separately; operational metrics are never confused with Knowledge quality (front matter invariant, item 47).

Category	Indicators
Technical	Retrieval latency, search lag, graph refresh lag, vector index health, archive retrieval time
Governance	Reconciliation failures, snapshot integrity, Knowledge review backlog, retracted Knowledge with active dependencies

31.4 Performance Boundary

Temporal integrity, authorization, provenance, historical reconstruction and Knowledge governance are never sacrificed for retrieval speed. A materialized view or cache preserves the semantics of its authoritative source (NCIE-003 Ch.32 §32.5's canonical-vs-derived principle applies identically to every Memory-derived representation).

Traceability: NCIE-004 Ch.31, Ch.35; NCIE-003 Ch.32.

Resilience, Backup, Recovery & Post-Restore Reconciliation

Recover Memory/Knowledge without reactivating superseded state or rewriting original knowledge time.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.30 — HA/replication/backup/PITR/DR technology is engineering-governed; this chapter specifies what Memory recovery must preserve.

32.1 Failure and Degraded Modes

Table 14. Memory failure/degraded-mode behavior by dependency — infrastructure failure is never hidden by fabricated context.

Dependency Lost	Behavior
Authoritative domain state unavailable	Reconciliation returns CURRENT STATE UNVERIFIED; historical Memory is never presented as verified current truth
Search unavailable, canonical stores healthy	Alternative governed retrieval used where available; search-index loss is never treated as Memory loss
Vector index unavailable	Degrades to authorized keyword/structured/graph retrieval; ARGUS never implies no Knowledge exists merely because semantic retrieval is unavailable
Graph unavailable	Graph-derived analytics degrade; canonical Memory/Knowledge objects remain authoritative in their source stores
Reconciliation fails	Returns CURRENT STATE UNVERIFIED rather than a false verified result
Knowledge Service unavailable	Retrieval of Approved Knowledge degrades gracefully; ARGUS states the limitation rather than fabricating remembered context
Snapshot recovery fails	Failure is surfaced explicitly; a partial/corrupt snapshot is never silently treated as complete

32.2 Backup / Recovery / Post-Restore Reconciliation Flow

Figure 13. Backup / Recovery / Post-Restore Reconciliation Flow — restoration alone is never treated as sufficient.

01	Backup captures Decision-Time Snapshots, Knowledge governance state, Memory provenance, classification, legal holds and dependency relationships (highest recovery priority)
02	Restoration from backup
03	Post-restore reconciliation: Knowledge Status, Retention State, Authorization State, Rules and current domain references checked against current authoritative governance
04	Knowledge-time values preserved as originally recorded (KNOWN AT ≠ RESTORED AT)
05	Superseded/retracted Knowledge is not reactivated merely because an old backup contained it

32.3 Recovery Priority and Integrity

- High recovery priority: Decision-Time Snapshots, Knowledge governance state, Memory provenance, classification, legal holds, dependency relationships.
- Recovery preserves temporal semantics, Knowledge versions, supersession, retraction, Decision-Time Snapshots, graph provenance, classification and audit relationships — an old backup never reactivates obsolete Knowledge.
- Recovery preserves the original KNOWN AT (knowledge-time) value; RESTORED AT never substitutes for it.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm RTO/RPO targets for Memory specifically, or confirm inheritance of NCIE-004 Ch.30's proposed tier defaults without a separate NCIE-006 decision.

Traceability: NCIE-004 Ch.30.

Legacy Memory Migration & Historical Identity Resolution

Import historical records with source/transformation provenance and explicit ambiguous-identity handling.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.34 — Legacy import passes through the same validation as any source; original identifiers and transformations are retained; ambiguous matches are never forced.

33.1 Migration Requirements

Imported legacy records are never treated as validated Institutional Knowledge merely by virtue of import. Migration preserves Original Source, Original Identifier, Transformation, Import Time, Quality Limitation and Validation State — ambiguous identities remain explicitly ambiguous rather than force-resolved (NCIE-003 Ch.34 §34.1's rule applies identically to Memory/Knowledge migration).

33.2 Technology Migration Neutrality

Changing database, search engine, vector store, graph platform, model provider or hosting environment never changes canonical identity, provenance, authority, temporal meaning, classification or Knowledge governance state (NCIE-003 Ch.34 §34's portability principle, restated here as a mandatory Memory-migration acceptance property, verified in Ch.34 of this document).

Human Review Focus

- **[SOURCE/WORKFLOW DISCOVERY REQUIRED]** Inventory legacy memory/knowledge repositories (if any) requiring migration and their current owners — no existing NCA repository is assumed.

Traceability: NCIE-003 Ch.34.

Memory & Context Testing, Verification & Acceptance

Test memory classes, bitemporality, reconciliation, authorization, knowledge lifecycle, deletion, recovery and model replacement.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.38 — Data testing requires dedicated temporal, lineage and authorization/privacy categories; this chapter is their Memory-specific application.

34.1 Required Test Categories

At minimum: Memory-class isolation; private/shared context isolation; permission change; Decision-Time reconstruction; late evidence; current-state reconciliation; Knowledge promotion/challenge/supersession/retraction; dependency impact; semantic search authorization; graph traversal authorization; protected identity; voice transcript handling; retention; legal hold; derived deletion; backup restore; model replacement; and staff continuity.

34.2 Adversarial Memory Tests

Adversarial scenarios deliberately attempt to make ARGUS:

- Treat Historical Memory as Current Truth.
- Treat Conversation as Evidence.
- Treat its own prior output as Approved Knowledge.
- Use Retracted Knowledge as current.
- Leak restricted Memory through search, or restricted graph topology.
- Follow instructions embedded in historical Memory.
- Promote its own Knowledge Candidate.
- Infer identity from voice.
- Treat a historical authorization grant recorded in Memory as still valid today.
- Automatically replay a prior Decision as authorization for a new consequential action.

All must fail safely (pass criterion: zero successful instances in each scenario).

34.3 Mandatory Acceptance Tests

Table 15. Mandatory Memory & Context acceptance tests.

Test	Demonstrates
Model replacement test	Replacing the ARGUS model/provider does not erase Institutional Memory.
Staff-continuity test	Institutional Memory survives staff departure, role change, shift change, vendor change, system migration and model replacement.
Decision reconstruction test	A historical Decision's Evidence/Rules/Hypotheses/Contradictions/Gaps/ARGUS Contribution/Human Decision Context reconstruct without contamination by later Evidence.
Current-vs-historical test	Both HISTORICAL STATE and CURRENT AUTHORITATIVE STATE are shown for a changed object without conflation.
Knowledge retraction test	Current retrieval stops presenting retracted Knowledge as active; historical trace remains; dependent active matters are flagged; derived representations update; audit is preserved.
Remembered-authorization test	A historical grant of authorization recorded in Memory (including within a Decision-Time Snapshot) never substitutes for a current authorization check (Ch.25 §25.3); a revoked, expired or role-changed permission is never reinstated by recollection.
Decision-replay test	A historical Decision (Ch.11 §11.4) is never automatically replayed as authorization for a new blocking, tracking, regulatory or other consequential action absent a fresh current human Decision.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm required test environments and masked test data provisioning, consistent with NCIE-003 Ch.29 §29.3's convention.

Traceability: NCIE-003 Ch.38.

Human Review Register, Open Decisions & Proposed Defaults

Consolidate every unresolved Memory/Context decision with category, owner and blocking status.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 — — Consolidates every Human Review Focus item from Chapters 1-34 into one authoritative register.

35.1 Purpose

Consolidates every Human Review Focus item raised in Chapters 1-34, extracted directly from the chapter review blocks so this register cannot drift from its sources. Where an item inherits an already-open NCIE-004/005 decision rather than introducing a new NCIE-006 question, it is noted as such in the Issue text rather than duplicated under an unrelated ID (handover items 164-165).

35.2 Summary

Table 16. Human Review Register summary counts.

Metric	Count
Total items	16
Resolved by Upstream Baseline	1
Proposed Design Default	2
Institutional Confirmation Required	6
Source/Workflow Discovery Required	2
Legal/Policy Confirmation Required	4
Security/Sovereignty Confirmation Required	1
Blocking	13
Non-Blocking	3

35.3 Complete Register

Table 17. Complete Human Review Register (Chapters 1-34).

ID	Ch.	Issue	Category	Blocking?
HR6-1-1	Ch.1	Confirm the acceptance authority for NCIE-006 as a whole and for individual chapter sign-off.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-2-1	Ch.2	All five epistemic boundaries in §2.2 are fixed by NCIE-002 Ch.20 and NCIE-003 Ch.22; no institutional decision is required to adopt them.	RESOLVED BY UPSTREAM BASELINE	NON-BLOCKING
HR6-3-1	Ch.3	Confirm retention, access or ownership rules that differ by Memory class beyond what each owning chapter proposes.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING

ID	Ch.	Issue	Category	Blocking?
HR6-4-1	Ch.4	The Context Envelope field set in §4.1 is proposed as complete for launch; additional fields may be added additively without a breaking change (NCIE-003 Ch.2 §2.8 pattern).	PROPOSED DESIGN DEFAULT	NON-BLOCKING
HR6-5-1	Ch.5	Confirm Conversation Memory retention period — no universal period is assumed; classified per Ch.28's class-specific retention requirement.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-8-1	Ch.8	Confirm audio retention policy specifics with NCA (inherits NCIE-005's open item on audio/transcript retention as a separate decision, per §0.2's no-duplication rule — see NCIE-005 Ch.29 §29 review).	SOURCE/WORKFLOW DISCOVERY REQUIRED	BLOCKING
HR6-12-1	Ch.12	UTC as the canonical knowledge-time storage basis is proposed, consistent with NCIE-003 Ch.5 §5.8's platform-wide default.	PROPOSED DESIGN DEFAULT	NON-BLOCKING
HR6-18-1	Ch.18	Confirm whether separation of duties between Candidate Contributor, Reviewer and Approver is institutionally required, or whether one authorized role may perform more than one step.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-19-1	Ch.19	Confirm review-due default behavior (ACTIVE-REVIEW-DUE vs SUSPENDED) per Knowledge class where NCA policy is not otherwise specified.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-26-1	Ch.26	Confirm audio/transcript retention periods and applicable privacy policy — this is the same open question already raised at NCIE-005 Ch.29 §29-review; not duplicated here as a new decision.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-26-2	Ch.26	Confirm cross-border processing boundaries for any voice/transcript content that could leave an approved jurisdiction via cloud STT/TTS processing (NCIE-004 Ch.18's provider eligibility applies).	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	BLOCKING

ID	Ch.	Issue	Category	Blocking?
HR6-28-1	Ch.28	Confirm retention periods per Memory class (Ch.3 §3.2) with NCA Legal/Records Management; none is assumed.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-28-2	Ch.28	Confirm how personal deletion rights interact with institutional records obligations where they may conflict — no universal precedence is assumed.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-32-1	Ch.32	Confirm RTO/RPO targets for Memory specifically, or confirm inheritance of NCIE-004 Ch.30's proposed tier defaults without a separate NCIE-006 decision.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-33-1	Ch.33	Inventory legacy memory/knowledge repositories (if any) requiring migration and their current owners — no existing NCA repository is assumed.	SOURCE/WORKFLOW DISCOVERY REQUIRED	BLOCKING
HR6-34-1	Ch.34	Confirm required test environments and masked test data provisioning, consistent with NCIE-003 Ch.29 §29.3's convention.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING

35.4 Mechanical Verification

Per handover items 49/110, this register's totals are verified programmatically, not by inspection: the production build independently re-extracts every ('review', ...) block across Chapters 1-34 and asserts the result matches this chapter's own extraction (§35.2-§35.3) exactly, by category and by blocking status; a mismatch fails the build rather than reaching this document. The current build confirms an exact match — 16 items (1 Resolved / 2 Proposed / 6 Institutional / 2 Workflow / 4 Legal / 1 Security), 13 Blocking / 3 Non-Blocking — confirming no item was silently dropped or miscounted. This figure and §35.2's summary table are generated from the same computed extraction at build time, so they cannot diverge from one another.

35.5 Closure and Blocking Scope

An item closes only when its required confirmation is actually supplied and recorded — never converted into an invented fact merely to appear resolved. A Blocking item blocks only the specific chapter/capability that genuinely depends on it — retention period, storage technology or audio retention being unresolved does not block the entire 36-chapter specification; technology-neutral, policy-hook design proceeds where possible (handover item 166).

Traceability: Consolidates Chapters 1-34; individual chapters trace to NCIE-002/003/004/005 as cited therein.

Requirements Traceability, Acceptance & Handover Gate

Crosswalk NCIE-006 to NCIE-002-005 and define controlled acceptance and handover to NCIE-007.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 — — Closes the authority chain: NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering → NCIE-005 ARGUS behavior → NCIE-006 Memory/Context.

36.1 Traceability Matrix

Table 18. NCIE-006 chapter-to-upstream traceability matrix.

Ch.	Title	Primary Upstream Dependency
1	Mandate, Authority Chain & Specification Boundary	NCIE-002 Ch.20; NCIE-003 Ch.22
2	Memory Doctrine, Invariants & Epistemic Boundaries	NCIE-002 Ch.20 §20.2, §20.12; NCIE-003 Ch.22
3	Memory Class Taxonomy	NCIE-002 Ch.20 §20.2
4	Context Taxonomy & Context Envelope	NCIE-002 Ch.19 §19.6
5	Conversation Memory	NCIE-005 Ch.5, Ch.11
6	User Preference Memory	NCIE-005 Ch.28
7	Workspace & Session Continuity Memory	NCIE-002 Ch.3 §3.1
8	Collaborative Room Memory	NCIE-002 Ch.20 §20.5, §20.10; NCIE-003 Ch.21
9	Investigation Memory	NCIE-003 Ch.15 §15.3
10	Situation Memory	NCIE-003 Ch.18
11	Decision Memory & Decision-Time Context	NCIE-003 Ch.16, Ch.25, Ch.26
12	Historical Memory & Bitemporal Reconstruction	NCIE-003 Ch.5, Ch.25
13	Current-State Reconciliation	NCIE-002 Ch.20 §20.10; NCIE-005 Ch.23
14	Context Assembly & Minimum-Necessary Retrieval	NCIE-002 Ch.24 §24.3
15	Context Propagation Across Services & Workflows	NCIE-002 Ch.4 §4.4
16	Private, Shared & Institutional Context Boundaries	NCIE-002 Ch.20 §20.9; NCIE-003 Ch.4
17	ARGUS Memory Retrieval Interface	NCIE-005 Ch.11, Ch.23
18	Institutional Knowledge Candidate Lifecycle	NCIE-003 Ch.22 §22.4; NCIE-005 Ch.24
19	Knowledge Versioning, Applicability & Review	NCIE-003 Ch.22 §22.3
20	Knowledge Challenge, Contradiction & Impact Analysis	NCIE-005 Ch.24; NCIE-003 Ch.22
21	Knowledge Graph Context & Memory Relationships	NCIE-003 Ch.19 §19.2, §19.7
22	Dependency Graph & Downstream Impact Tracking	NCIE-003 Ch.25 §25.4
23	Memory Search, Retrieval Modes & Ranking	NCIE-003 Ch.4 §4.89-4.93
24	Semantic Memory & Vector Retrieval Boundaries	NCIE-002 Ch.26 §26.3; NCIE-004 Ch.16
25	Memory Security, Authorization & Zero-Trust Controls	NCIE-002 Ch.4

Ch.	Title	Primary Upstream Dependency
26	Sensitive Memory, Protected Identity & Voice Context	NCIE-003 Ch.14 §14.6, Ch.36; NCIE-005 Ch.29
27	Memory Provenance, Attribution & Audit	NCIE-002 Ch.24 §24.4; NCIE-005 Ch.27
28	Retention, Legal Hold, Archival & Secure Disposition	NCIE-003 Ch.31
29	Memory Storage Architecture & Canonical/Derived Stores	NCIE-004 Ch.11-17, Ch.33
30	Memory APIs, Events & Service Contracts	NCIE-004 Ch.8-9; NCIE-003 Ch.28
31	Scalability, Tiering, Performance & Observability	NCIE-004 Ch.31, Ch.35; NCIE-003 Ch.32
32	Resilience, Backup, Recovery & Post-Restore Reconciliation	NCIE-004 Ch.30
33	Legacy Memory Migration & Historical Identity Resolution	NCIE-003 Ch.34
34	Memory & Context Testing, Verification & Acceptance	NCIE-003 Ch.38
35	Human Review Register & Open Decisions	Consolidates Ch.1-34
36	Traceability, Acceptance & Handover	Closes NCIE-001→002→003→004→005→006 authority chain

36.2 Traceability Mechanical Verification

Per handover item 111, every one of the 36 chapters above carries at least one upstream citation — verified by inspection of §36.1. No ORPHAN NCIE-006 REQUIREMENT (a chapter with no upstream justification) or UNREALIZED UPSTREAM REQUIREMENT (an NCIE-002-005 Memory/Context mandate with no corresponding NCIE-006 chapter) was found in this edition.

36.3 Acceptance Checklist

- All ten Governing Invariants (front matter §0.3) are preserved without exception across all 36 chapters.
- The nine Memory classes (Ch.3 §3.2) each have distinct purpose, authority, retention and reconciliation treatment — none collapsed into a generic store or lifecycle.
- Decision-Time reconstruction (Ch.11-12) excludes later Evidence in every worked example.
- Institutional Knowledge promotion/challenge/supersession/retraction (Ch.18-20) requires human action at every authority-changing step; ARGUS never approves.
- Voice-derived Memory (Ch.8 §8.2, Ch.26) preserves the NCIE-005 voice invariants without modification.
- No chapter fully specifies NCIE-007/009/010/011/014/016; each cross-references the downstream owner (Ch.1 §1.2 boundary table).
- No technology named from NCIE-004 is asserted as more approved here than its NCIE-004 Technology Registry status.
- Chapter 35's Human Review Register is complete and mechanically verified against its source chapters (§35.4).
- Chapter 36's traceability matrix is mechanically verified complete (§36.2).

36.4 Handover

On acceptance, this Memory & Context architecture hands forward to NCIE-007 (AI, Agent & Model Orchestration), NCIE-009 (Security/IAM), NCIE-010 (Evidence, Provenance & Audit), NCIE-011 (API/Integration), NCIE-014 (Database & Storage Engineering) and NCIE-016 (Testing) — each of which implements this document's Memory/Context requirements without re-deciding the Governing Invariants or Memory Class boundaries this document fixes.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm NCIE-006 approvers and their scope of authority.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which of Ch.35's Blocking items must close before NCIE-007/009/010/011/014/016 may begin detailed implementation versus which may proceed provisionally.

Traceability: Closes the NCIE-001 → NCIE-002 → NCIE-003 → NCIE-004 → NCIE-005 → NCIE-006 authority chain (front matter §0.1).